

Double-Binds and Anamorphism in *FEZ*'s Reorientation

In 2012, Polytron released the experimental 2D puzzle-platformer *FEZ*, which begins with its player character, Gomez, encountering an extradimensional being known as the Hexahedron. It reveals to Gomez that he is actually a 2D organism in a 3D world and blesses him with the titular fez, which affords him perception of the third dimension, but only by fragmenting his world into four 2D faces. This exchange between realities also leads to the fragmentation of the Hexahedron, thrusting Gomez and the player into a quest that will rely on their ability to utilize the power of the fez to navigate the game's world and attempt to stabilize reality. This power, which takes the form of the game's primary mechanic, *reorientation*, allows Gomez to rotate his 2D vision between the four faces of 3D space. By shifting perspective, the actual positions of objects in Gomez's reality are changed accordingly; the player and Gomez can use this to navigate between different sets of possible action when they seemingly lack a means to progress or a clear goal to progress towards. However, the shifting of one perspective to the foreground necessitates that certain aspects of the previous view are temporarily lost to the background. Thus, as the player and Gomez navigate this fragmented and non-linear world, cut up into four perspectives, they must repeatedly traverse the same environments to exhaust their possibilities. This necessity of repetition, driven by an anamorphic structuring of reality via perception, is extended by later revelations of the in-game languages and still unsolved final puzzle, pointing towards a potential inexhaustibility of action directly mediated by one's relative perspective.

FEZ begins in Gomez's homeworld, Villageville. After gaining control of Gomez, the player and him receive a letter from the old man, Geezer, who awaits them at the top of the village. The letter informs the two to go find Geezer because "Today is special day," (Polytron).

This section, wherein the player must control Gomez to reach the top of Villageville, operates as a standard sort of tutorial for the game's core platforming mechanics; a simple goal is established: reach the top of the village by learning to move, jump and grab ledges. As many video games go, an introductory tutorial allows the player to get used to their primary means for exploring and navigating the goal structures of the world. This kind of learning, calibration to a new system or *mechanical set*, clearly aligns with Gregory Bateson's category of *Learning II*, wherein a player makes "a corrective change in the set of alternatives from which choice is made," (310). To take up playing *FEZ*, the player must first learn the set of options *for* platforming; they learn to overcome obstacles or achieve goals, such as climbing to the top of Villageville, by first learning to utilize their mechanical set of movement. More specific and directly goal-related problem solving similarly aligns with *Learning I*, a "*change in specificity of response* by correction of errors of choice within a set of alternatives," by moving and jumping at the right time and place to successfully reach platforms (310).

The mechanical set learned in *FEZ*'s tutorial is fairly standard for conventions of a 2D platformer. The simple options of moving and jumping to reach a given platform have been genre staples since its conception. For an initiated player of 2D platformers, these conventions permit what Bateson calls "set learning" in regards to *LII*, where, for problems that rely on similar mechanical sets, "there [is] a carry-over of learning from one problem to the next," (312). The conventions permit a player of *FEZ* who has previously learned-to-learn inside a 2D platformer to fairly comfortably navigate the game's context for action. This comfortability of a familiar context is notably reflected by the overall tone and atmosphere established by the characters inhabiting Villageville. One of the first villagers the player can interact with states that his favorite shape "is square! Not cube, that's for sure! Because there is no such thing,"

(Polytron). The next tells Gomez and the player that she “loves village” and that she’s lived here her whole life, (Polytron). These snippets of dialogue in quick succession may inform the player that a familiar 2D perspective has in some ways confined these villagers' view of their world. This relation nicely reflects the previously set-learned player’s own comfortability within the context of a conventional 2D platformer. However, an old man just below the village’s schoolhouse tells Gomez, and the player, that “Reality is perception. Perception is subjective,” gesturing towards the fact that a familiar perspective, despite its comfortability, is not the only way of engaging with reality (Polytron).

This claim of a perspectival organization of reality, as well as the comfortable limits of perception in Villageville, are both neatly grounded by Abraham Moles’s studies on the *apperceptual limit* of humans as receptors of information. Regarding perception’s general confinement to familiarity, Moles notes that a subject’s *a priori knowledge*, or previous learning, directly relates to their apperceptual limit, or ability to make sense of their current situation and context for interaction with reality (93). An inhabitant of Villageville could then be understood as being simply unable to make sense of information that exists outside of their 2D frame of reference, namely, 3D space. This inability to perceive outside of a 2D context for interaction, or mechanical set, shields them from the potential uncertainty of their apperceptual limits, allowing them to sit comfortably within familiarity, much like a player accustomed to conventional 2D platforming mechanics.

These comfortable apperceptual limits, which allow the villagers to navigate their day-to-day lives within the village, could be described as “good-enough” frames of reference. Villagers lack immediate exigency to reach beyond this limit and to learn new contexts or ways of seeing. However, this exigency could be produced by what Marie-Laure Ryan calls

“productive conflict,” which she considers essential to generating action within narrative. In order for a character to feel motivated to act within a narrative, there must be some sort of driving force: a separation between what is desired by or expected of a character and the actual world they occupy. Ryan conceptualizes this essential dissonance through the relative frames or perspectives of *model-worlds* which form subjunctive propositions of what the world could or should be. In the context of model-worlds, productive conflict is generated by the moment when “some of the propositions of a private world” or a subjective frame of reference, “become unsatisfied in the actual world” which causes the narrative universe to fall into “a state of conflict,” (733). In the beginning of FEZ, none of the villagers, including Gomez, experience a kind of distance between their models and the actual world which would necessitate a shift in their familiar perspectives. The alignment between their perception and reality is good-enough that the 3D world holds no relevance. However, today is a special day and Gomez’s knowledge of reality will soon be expanded, fragmenting reality itself and driving his model-worlds away from the real one; he and the player both will then be required to make use of this new knowledge in order to put the world back together.

As the player completes *FEZ*’s platforming tutorial, aligning a model with the game’s actual world, to meet Geezer, Geezer proclaims, “Adventure is ready! It’s Gomez time!” (Polytron). After a couple seconds of waiting around for Gomez time, the shape of a hexagon forms in the sky above and begins to rotate, breaching the 2D perspective it formed within. A bright light engulfs the screen, transporting Gomez to a new environment and the shape, properly known as the Hexahedron, takes on its actual form, with gold tetrominos lining its exterior. It speaks to Gomez in an unknown language expressed visually by sequences of patterned squares and verbally as a seemingly incoherent *noise*. Noise, to Moles, is interference in the receptor’s

capacity to discern the intent of a message: noise is “*a signal that the sender does not want to transmit,*” (79). While the Hexahedron fully intends to express these signs to Gomez, Gomez’s inability to understand them provided his, as well as the initial player’s, apperceptual limits, turns this signal into functional noise, skewing its intent. However, this noisy message still denotes a lack inherent in Gomez’s apperceptual limit, providing a meta-message that signals to Gomez the existence of a different mode of perception. A ray of light is cast down onto Gomez, pulling him upwards into the air below the Hexahedron, all while this novel language falls down onto him as well. The message that falls onto Gomez, when deciphered through the discovery of an in-game cipher key, actually states, “You dummy, the world is in three dimensions,” flattening the seemingly distant intent of the noisy message and the salient meta-message (Polytron). While Gomez is held in the Hexahedron’s revealing light and language, he and the camera begin to rotate along with the Hexahedron, and light is thrown onto the third dimension. The formerly dissimulated background noise of the third dimension, that once bore no relevancy for Gomez, is dragged to the foreground of perception through the signal of a possible signal.

After the truth of reality is revealed and the Hexahedron’s spinning winds down to a halt, it mentions to Gomez that it “kind of thought maybe this would not work because of your weird head,” since Gomez’s head is a two-dimensional form within a three-dimensional reality. However, the Hexahedron assumes everything is fine and disappears, where another ray of light is cast down onto Gomez. The titular, three-dimensional fez falls down through the light, rotating along the axis of the third dimension, until it lands squarely atop Gomez’s 2D head. Gomez does a little joyful hop, the Hexahedron reappears, and the player is prompted to use either of the triggers on a controller, or the A and D keys on the keyboard to “change [their] perspective,” (Polytron). Here, the player acquires the game’s most central mechanic: *reorientation*; with

reorientation the player can shift their two-dimensional feedback, likely limited by Gomez's two-dimensional form, to the neighboring left or right faces of the three-dimensional world, playing with their contingent mechanical set or possibilities for action.

This moment of learning, wherein Gomez and the player are now able to see the context of potential 2D platformer contexts, is well classified by Bateson's final relevant category of learning, *Learning III. LIII* is defined as a "corrective change in the system of *sets* of alternatives from which choice is made," wherein an organism's habitual understanding of reality, and thus their means of perceiving contexts in the modality of *L2*, is modified (310). The apperceptual limit of two-dimensions, which establishes the context for the game's mechanical set, is revealed by the Hexahedron to only be *a* familiar system by which the context for action is discerned. Bateson notes that it is often incredibly difficult and dangerous for an organism to attempt this level of learning, yet it may be accomplished by having the organism confront "contradiction" within the structure of their learned behavior, (303). He shortly after clarifies that elsewhere he has "called these contradictions at level II 'double-binds.'" which could be thought of as a paradoxical overlay of two contradictory contexts for perception (304). In Gomez's case, as a 2D organism with a 3D hat, his current state is immediately implicated in an interdimensional double-bind; this contradiction is resolved at the level of *LIII* via the mechanic of reorientation, which allows for 2D embodiment within a 3D reality. To see beyond two-dimensions in this case is not simply the ability to adapt to a new context, or learn a new mechanic, but rather, it is a kind of learning that, through the resolution of a contradiction, enables the player and Gomez to redefine how they determine the context of their possible actions. Their previous perceptual system is now synthesized into a manually operated figure/ground distinction, where parts of the

2D world can be dissimulated, or recontextualized as noise, in order to bring about changes in their immediate context.

However, as the player follows the game's instructions and tests out this new mechanic, shifting their perspective relative to Gomez's position, the Hexahedron begins to rotate at an increasingly rapid speed, until fracturing into numerous cubes. The camera continues to rotate without the player's input as various forms of static and ASCII symbols clutter the screen, until the game finally freezes and restarts, bringing the player back to the title screen. Though, the menu is different this time; the title "FEZ" begins to glitch and fragment at three different angles, colored red, green and blue, the colors of the human eye's photoreceptors, gesturing towards the fragmentation of the player's, as well as Gomez's, perceptual apparatus. Much like this perceptual fragmentation, *LIII*, according to Bateson, holds the potential to destroy the organism's conception of self-hood, inasmuch as its "self" will take on a sort of irrelevance," once it can act in terms of "the context of contexts," (320). The "self" then, an aggregate of consciously or unconsciously learned contexts for interaction, is revealed to be the modifiable system that binds these contexts together. The fragmentation that occurs as a result of Gomez's enlightenment then, literalizes the cognitive effects of *LIII*; through the resolution of a double-bind, his habitual perception of the context for action is resolved to include a higher dimension, the context of contexts, which reveals the fragmentary nature of *his* reality and shatters reality itself.

When the player reopens the game, Gomez wakes up in his room just as he did when the game first started; the difference this time is that he now wears the fez, and can thus reorient. When Gomez leaves his room he's greeted by a small extradimensional deity, Dot, who informs Gomez that something went wrong during his encounter with the Hexahedron. Dot tells Gomez

that the fragmented pieces of the Hexahedron must be put back together in order to prevent the universe from collapsing. It was shattered into 32 cubes: 17 that still hold their full shape and 120 cube-bits, which form a full cube when 8 are gathered; this task of collection forms the game's primary and ultimate goal. This ultimate goal could be thought of as a primary model-world that generates productive conflict within Ryan's conception of narrative. Due to the fracturing of reality as a result of Gomez's encounter with the Hexahedron, he and the player are now tasked with putting the world back together. They are thrust outside the confines of the familiar Villageville as a result of being thrust outside of the confines of their previous a priori perception. The achievement of this ultimate goal and a return to equilibrium will necessitate the use of *LIII*, or the mechanic of reorientation, to maneuver through a world of shifting contexts.

Reorientation, as a mechanic, is notably different from the other core mechanics of the game; instead of operating *within* the context of action, it directly manipulates the possibility space *for* mechanical action by acting upon the context of contexts. In line with Ryan's understanding of *state and event propositions*, the former being the set of truth values about instances within the storyworld, and the latter being the processes that effect a change in state, such as a mechanic in a game system, reorientation operates as an event proposition that deliberately skews state propositions through a manipulation of *feedback* (718). A game's feedback, according to Cardona et. al, is the perceptual stimuli that informs a player of the game's goals and possibility space (4). By shifting the camera, the relation of the in-game objects to the perceptual apparatus of feedback is altered, changing their actual state propositions in relation to Gomez and the player; the possibility space of reality is informed by perception and is thus synthesized through a subject/object relation. This proposition implicit in reorientation is further grounded in Moles's note on a receptor's intentionality in separating signal from noise:

when parsing signals from an environment or object, “the transmitter as an individual does not exist; only the receptor remains,” where noise simply becomes “*a sound we do not want to hear*,” (79). By dissimulating certain relations between Gomez and his environment, by shifting the more abstract relation of feedback, reorientation allows the player to receive different signals and dialectically construct a different state or context for action.

The first platforming puzzle that Gomez and the player face once beginning their quest functions as a simple tutorial for how to reorient in order to achieve a goal or close the gap between model and actual world. They enter an environment consisting of a single tower, with climbable vines snaking their way up it. From the initial perspective, only a small portion of these vines are accessible, which requires the player to reorient to the right once to follow the vines upwards and climb higher. Here, once the player has *reached* the end of their possible action within this perspective, they must reorient to recontextualize their relations to the world and establish a new goal. By climbing to the highest visible point on a set of vines, the player achieves what Cardona et. al would consider an *imperative, to reach*, goal (4). An *imperative goal* in this model is a single sub-goal in a hierarchy of parts that makes up the whole of a game’s *ultimate goal*; thus, in Ryan’s model, imperative goals could be thought of as a sort of sub-model-world which brings model and actual worlds *a bit* closer into alignment. The player must repeat this *reach, reorient* pattern a few times over until they find the first cube-bit of the room, establishing an imperative goal more directly related to their ultimate goal. This pattern continues until the player reaches the top of the tower and collects 8 cube-bits, rewarding them with a full cube, where they can then reorient to find a door that will lead them into the next room, nudging their model world a bit further along course. Reorientation’s effect upon the receptor’s contingent goal structure reveals that while the total model-world of FEZ may remain

constant, the parts that make up this goal, its form, is continually recontextualized by how Gomez and the player are oriented. And in fact, the gap between model and actual world can only be closed through a continual reorientation of the player's relation to it, by establishing new imperative goals via perceptual shifts.

The contingency of the player's relation to ultimate goals, mediated by the state propositions of feedback and thus their orientation towards the actual world, finds a clear explanation in Ryan's understanding of *intention-worlds*, which structure action according to the immediate relation of model and actual worlds. According to Ryan, an intention world is defined when "a character commits himself, privately or to witnesses, to reaching a certain target by following a certain course of action," (725). These worlds operate as synthesis of a character's knowledge-world, or their understanding of reality as conditioned by their orientation towards it, and how their model-world directs action through it. Reorientation literalizes this idea; by bringing about an alteration of feedback, and thus a change in foregrounded state propositions, it reveals a different set of imperatives and reorganizes the player's ability to reach their ultimate goal. The state of the narrative, the distance between model and reality, and the significance of action itself, becomes directly mediated by the intended filtering between signal and noise, i.e. perception.

Each of the imperative goals encountered in this introductory area, such as platforms, cubes/cube-bits and doors, hold unique structural significance within the context of FEZ's narrative, given they each relate to the ultimate goal in different ways. The cubes and cube-bits operate simply enough: as the player reaches each one, the distance of how many more they need to reconstruct the Hexahedron goes down, directly closing the gap. Due to their direct proximity to the ultimate goal, these could be thought of as a *definitive imperative goal*, informing the rest

of the player's imperative structure. However, the significance of platforms, which is guided by the pursuit of either cubes and cube-bits or doors, which lead the player to a new room with new goals (and will be discussed later), is more complicated. Their shifty significance is best highlighted by a later puzzle involving another tower that holds the shape of a donut; on two sides, this tower is the shape of a square with a circle cut out of its center, and on the other two, a tall rectangle. In the center of the circle is a cube-bit that structures the player's action, yet none of the climbable vines, a form of the platform imperative, afford a clear path to the center of the donut and this definitive goal. In the initial moment of encounter, their narrative significance is an ambiguous signal.

On the side opposite to where the player first enters the room, they can find the only immediately reachable vines, which at first connect to nothing and seemingly lead nowhere. However, if the player climbs them and reorients to the right once, the distance between those vines and newly foregrounded vines that climb further up is dissimulated according to perspective, allowing the player to continue climbing. While mechanically similar to the *reach*, *reorient* pattern encountered in the previously detailed tutorial for reorientation, the narrative significance of these vines remains ambiguous *until* punctuated by a contextual shift. This shifting meaning of action is only further exemplified by the final puzzle on the donut which requires the player to move to a specific position on a set of vines before reorienting, lest they be cut off from their goal of reaching the cube-bit by their new perspective. Here, by shifting between contexts of *LII*, reorientation functions as what Bateson calls a *context-marker*, which affects "the manner in which the stream of action and experience is segmented or punctuated," (310). In the case of these vine puzzles and the *ambiguous reach*, *reorient* pattern, the player must input an *action-syntagm*, a selection and combination of motions within a context, to reach

a seemingly meaningless position, and punctuate towards a meaningful context, with a shift to the right, in order to articulate a motion with positive narrative value. This shifty potentiality of the narrative value of an action is also resonant with Jacques Lacan's understanding of punctuation, wherein "a sentence closes its signification only with its last term, each term being anticipated in the construction constituted by the other terms and, inversely, sealing their meaning by its retroactive effects," (682). Here, the player's action-syntagm anticipates the imperative goal of *reaching* the top of the vine, which enables them to correctly punctuate their action by connecting it to a new context, one that houses another possible imperative goal. In general, each action articulation achieves narrative relevancy only retroactively through the contextual shifts or punctuation that link imperative goal to imperative goal to ultimate goal; the *ambiguous reach, reorient* pattern simply literalizes the continual contextual shifts that chain together the total stream of goal-directed and meaningful action.

The way in which context and goal structure define and redefine the articulation and meaning of the player's action extends beyond just single rooms when examining the imperative goal of reaching *doors*. Most rooms in *FEZ* house definitive imperative goals, whether a cube or cube-bit or deeper secrets that play a role in the game's second ending, and once these definitives are completed, the player's map marks that room with a golden border, letting them know it has been exhausted of immediate narrative value. Once all immediately possible definitive goals in a room have been achieved, doors allow players to move on to another set of imperative goals within the game's chain of action. Exploration of the game's map then forms a larger action-syntagm, a *map-articulation*, wherein each room, as a term in the chain of exploration, gains its narrative value through its relation to shifts in context or the player's orientation. However, given reorientation's fragmentary nature, the map itself takes on a non-linear,

networked format, where one room-node may house several doors that branch out into any of the game's four directions, or even up or down if the room is large enough vertically. This form requires the player to repeatedly traverse and rearticulate the meaning of previously exhausted rooms in order to exhaust the total map of its imperatives.

FEZ's map is divided into four primary sections: Natural, Industrial, Sewer and Cemetery. The central nodes of each environment are linked together through a form of fast-travel, a Warp Gate, which provides a portal to the other environments depending upon the player's current orientation. Each of these environments could be thought of as a discrete sub-section of the game's map, and thus, a sub-section of imperative goals. The nodal nature of the map, fragmented by reorientation, requires the player to choose only one direction to progress in at each node, whilst leaving the other potentialities for action as dissimulated noise for the time being. This anamorphic shifting of possibility that enables some meaning and disables the rest, falls closely in line with the principle of *exclusion* which Lacan finds essential to the interaction of subject and object. Specifically in reference to Lacan's work on the relation of the gaze and the anamorphic painting *The Ambassadors*, Maria Scott notes that he believes the "painting dramatizes the working of human vision, and analogically, the constitution of subjectivity," given that both vision and an embodied semiotic structure require the subject to exclude some information to make sense of any (328). Through the traversal of *FEZ's* map, the player manipulates and reads the shifting feedback of their environment, enabling and disabling specific imperative goals and map-articulations; yet the excluded terms do not simply disappear, they simply await the use of reorientation to bring them into alignment.

Once the player arrives at the end-point of an imperative network, they will find a Minor Warp Gate, a one-way portal which sends the player back to the central Warp Gate of their

current environment, nudging them to articulate the possibility space once more. Upon the arrival at an end-point, that specific map-articulation is punctuated, sealing the meaning each input and room had within it. However, if their connected nodes aren't yet exhausted, these rooms still await exploration. The player can open their map to determine which parts remain unexplored, and must then rearticulate the same environment to find another end-point, placing each input and room within a different syntagm. This necessity of repetition, being driven by the anamorphic principle of *exclusion*, is reflected in Lacan's theory of *repetition automatism*, wherein "a hole opens up that is constituted by a certain caput mortuum of the signifier" as a result of the restraint of possibility each term in a chain carries with it, which "suffices to make it depend on absence, obliging it to repeat its contour," (38). As the player articulates a non-linear possibility space in a linear, syntagmatic form, certain possibilities are naturally mutually exclusive, forcing a rearticulation, and a reconstitution of each term's meaning in relation to the total action-syntagm of play. Each room's narrative significance, and thus its meaning as a virtual object, is continually redetermined by how the player chooses to orient themselves towards it as they loop through the same environments in order to achieve the game's ultimate goal.

Throughout their quest, doors will open underneath the central Warp Gate corresponding to the player's cube progression in powers of two, allowing them to observe the history of an ancient civilization central to the game's lore. Once 16 cubes are acquired they're allowed to access the current state of that civilization, and inside that 16 cube door is the 32 cube door, the number of the game's stated ultimate goal. When the player completes the network of imperative goals scattered through FEZ's map through the articulating the acquisition of 32 cubes, the possibility space of meaningful action within the game becomes seemingly exhausted. At this point, each narrative enigma locked behind progress markers have also been revealed; the game's

story seems to have come to a close. With the ultimate goal in sight, Gomez and the player travel through the 32 cube door, through an interdimensional star gate inside it, and arrive in a blank white room, where Gomez's color scheme is now displayed in negative. A black hexagon, the frame of the Hexahedron, pops into view above Gomez and begins to rotate as the 32 cubes fill the frame. It stops rotating once the last cube has been inserted, however, the supposedly final 32 cubes, which are even noted as 100% progress in the player's save file, are revealed to only fill out half of its form. As the Hexahedron begins to rotate again, something seems to be wrong: it continually flashes between the two perspectives of its completed tetromino-lined form and its current half-filled state, all while noise and scattered ASCII symbols once again clutter the screen. After a few rotations, the game restarts, just like the first encounter.

The game restarts once again in Gomez's room, and the player and Gomez, again, climb to the top of Villageville; the game's action-syntagm recurses and the first imperative goal is recontextualized as the last, shifting its meaning through diachrony. As the player retraces their first steps, the camera zooms in on Gomez while the game's pixelated graphics begin to smear and the quality of the game's audio begins to decline. The zoom stops once Gomez's pixel body has become nothing but a tetromino of a white square with a smaller red pixel atop it; the parsing of signal from noise reaches a specificity via exclusion *ad absurdum*. The red square, the fez, falls off, and the camera enters it, revealing an entire red wireframe map inside of this single pixel. Above this wireframe forms yet another rotating hexagon, which turns to reveal a hole on its initially occluded side; as the camera travels through it, this hexagonal hole recurs in a potentially infinite regress, calling to mind the caput mortuum of the signifier and the necessary exclusion which sustains each and every perspective yet enables new modes of seeing. This recursive visual sequence continues until finally zooming out from a black screen, out of

Gomez's eye, where the end credits sequence finally plays. After this, the player is given the ability to begin New Game +, where, akin to the visual logic of this recursive and looping end sequence, they and Gomez will repeat their paths through the world guided by new signals and new goals. As Bateson notes, the purposive nature of goal-driven consciousness reveals "only arcs of circuits, cut off from their matrix by our selective attention," and the guidance of a different goal will thus generate a different sort of perception, with different exclusions (451). These final systems and puzzles will redefine the game's seemingly exhausted possibility space and negate the notion of a finalizing frame of reference or a point of arrival for meaning.

When the player opens up their NG+ file and reencounters the Hexahedron, Gomez is once again provided a signal of a possible signal; however, this time Gomez is given the First Person Glasses, which ironically, allows the player to view the world directly through Gomez's eyes, what should be a familiar mode of perception to Gomez. This creates a sort of metatextual double-bind, where the contradictions of the player's own embodiment of the player character are called into question. This mechanic, operating still as yet another means of feedback manipulation and recontextualization, is an essential tool for finding the second half of the Hexahedron and completing the game's second ultimate goal. This ending requires that the player not only collect all 32 cubes, but to find their excluded back-halves, 32 anti-cubes, as well as 4 heart-cubes by solving each and every puzzle hidden within the map. The First Person Glasses, in this case, function as a kind of inter-semiotic repetition automatism; the player can now parse information that was previously excluded or occluded, even within the reality transforming perspective of reorientation, and discover new imperative goals. The fact that Gomez's secondary enlightenment informs the player of details that were, quite literally, right in front of them, akin to the map nested inside of the red pixel, points towards the very

inexhaustibility of an anamorphic perception once confronted with its contradictions, and thus an inexhaustible meaning of action oriented by it.

At some point, during their second interrogation of the game's environment and puzzles, the player will eventually stumble upon a machine which outputs tetromino symbols in accordance with the player's input. This machine works as a cypher which enables the player to translate vertical sequences of tetrominoes placed throughout the world, previously hidden in plain sight, directly into action-syntagma. These tetromino symbols correspond directly to the central mechanics of movement, jumping, and reorientation, allowing the player to read feedback and directly translate it into action. However, in order to read some of these, the player must utilize Gomez's first person perspective, whilst also literally reorienting their own vision to read all of them, as each vertical sequence must be rotated ninety degrees to be translated properly; the structure of the player's own engaged perceptual apparatus and how it might be related to the virtual world becomes a part of the game's ruleset. In this case, the manipulation of feedback to parse a signal reaches an absurd point of complexity, where the player is all at once translating a symbolic system, a virtual orientation and their actual physical orientation in order to understand an imperative goal.

This *complex anamorphism* reveals the flexible limits of the human receptor's ability to extract signal from noise and to construct a possibility space for action, where action becomes dependent upon the seemingly inexhaustible possibilities of compounded perception. Like the past expansion of perspective and meaning-making proposed in *FEZ*, this moment relies upon the superposition of multiple seemingly contradictory contexts for perception. Bateson describes the double-bind, that which may generate the ability to learn at level III as "some sort of tangle in the rules" for parsing signal from noise whilst a subject engages with reality, (272). By

imposing several seemingly contradictory rulesets for making meaning and exposing their narrative contradictions, such as needing glasses to see from their in-game position or being able to manipulate their body in real and not virtual space, the player is able to creatively synthesize these contradictions and learn at LIII. The rules of *FEZ* quite literally teach the player the ways in which their deeply habituated rules for perception might be played with through a complex of contexts. Yet as this expanded possibility space of NG+, which forms as a result of complex anamorphism, still seemingly approaches a clear conclusion and exhaustion of meaning, the ability to properly orient the final meaning of the player's action will be complicated and potentially stumped by the game's final puzzle.

This final tetromino sequence, named after its central object, the Black Monolith, has still, fourteen years after the game's release, yet to be solved by any means other than brute force. During the first week or so of the game's release, players engaged in the online community were suggested to solve the solution through random attempts, given that no one could figure out where to look for the clues or rules for solving it. Now, the solution exists online, accessible to any player looking to complete the game and arrive at the point of 209% completion, yet no one understands how this solution might be arrived at organically through observation and play. Since having been solved fairly shortly after the game's release, the Black Monolith sequence has been read back into almost any seemingly relevant information inside or outside the game. None of these hypotheses have been stated to be correct by the developers and all of them rely on the sequence itself to arbitrarily and retroactively produce a signal. In light of this informational blank, the total sequence of play and action within *FEZ*, or the ability to arrive at its so-called ultimate goal which orients and solidifies the game's structure, is flaunted as nothing but an arbitrary confinement of the shifting play of meaning. When the final link in the action-syntagm

is a groundless form capable of being repeatedly overlaid onto noise through complex anamorphisms, like reading the “external” information of game’s release date back into the puzzle, the total narrative meaning of gameplay remains a perpetually shifting substance. The player’s search for a correct ruleset for making sense of information, through the combination of seemingly contradictory perspectives like ones which leak out into the metatextual digital sphere, continually and retroactively reshape the arc of the Black Monolith and the ultimate meaning of *FEZ*’s ludonarrative.

The pursuit of an impossibly final context for rule construction carried out by players attempting to make sense of the Black Monolith closely resembles an experiment Bateson himself commented on in respect to a porpoise subjected to continual double-binds. The porpoise was placed in an environment it was trained to perform a particular action in, wherein it was initially rewarded for performing the familiar action. However, after being taken out and placed back into the same test environment again, it assumed the context for its interaction with its experimenters stayed the same when the experimenters were in fact playing a trick on the porpoise: each time the porpoise was placed in the same environment, it would only be rewarded for acting according to a different and novel context. At first, “the experience of being put in the wrong” in regards to her context for interaction with another mammal was incredibly disturbing to her; however, “between the fourteenth and fifteenth sessions, the porpoise appeared to be much excited” and on the fifteenth session “she put on an elaborate performance including eight conspicuous pieces of behavior of which four were entirely new — never before observed in this species of animal,” (Bateson, 277). Through surviving the stress induced by the experience of continual double-binds, the porpoise discovered how to creatively augment her own rules for context creation. Much like the porpoise, a player of *FEZ* must continually repeat the same

environments over and over again, abiding and constructing new rules for their perceptual apparatus each time to creatively solve the game's contextual puzzles. The Black Monolith itself forces the player into the same position as the porpoise wherein no definitive point of arrival for meaning exists; rather, through the creative process of continually expanding and rewriting their rules of perception, the player experiences an autotelic joy in their potentially inexhaustible capacity for orientation towards reality.

Through presenting the player with a series of double-binds throughout play which increase in complexity as they approach the second and indeterminate ultimate goal, *FEZ* lures the player into learning how to learn to learn. By the end, spaces for interaction both inside and outside the initial context of the game become a seemingly inexhaustible possibility space for the ultimate narrative meaning of their action. As these continually expanded contexts for perception are overlaid on top of one another in a complex anamorphism, the player learns how to play with the rules of perception and find creative solutions through the resolutions of apparent contradictions. The process of free play is revealed to be precisely the ability to creatively extract signals from noise through the manipulation of one's orientation towards the world around them; as Bateson notes, "All that is not information, not redundancy, not form and not restraints – is noise, the only possible sources of new patterns," (416). In order to find new ways of being and paths towards one's desires in a reality that seemingly contradicts them, all one may have to do is simply change their perspective, transform noise into new signals, and to find joy in the process of play that enables their ability to make sense of the world.

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